

Qatar-1b

R-0595

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_240418 · created 2026-07-25T00:55:24+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460418.9166 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.02
Transit depth (Rp/Rs)^2	3.66 +/- 0.78 [%]
Orbital Inclination (inc)	81.98 +/- 0.95
Airmass coefficient 1 (a1)	0.996 +/- 0.013
Airmass coefficient 2 (a2)	0.0013 +/- 0.0085
Scatter in the residuals of the lightcurve fit is	1.34 %
Best Comparison Star	#3 - [433, 321]
Optimal Aperture	3.49
Optimal Annulus	10.06
Transit Duration (day)	0.0584 +/- 0.0092

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.191 $\pm$ 0.02 vs 0.1463 (deviation 2.23 σ)
Duration fit vs published	0.0584 d vs 0.0692 d (deviation 1.17 σ)
Mid-time O-C	2.1 min
Depth SNR	9.6
Residual scatter	1.34 %

Light curve

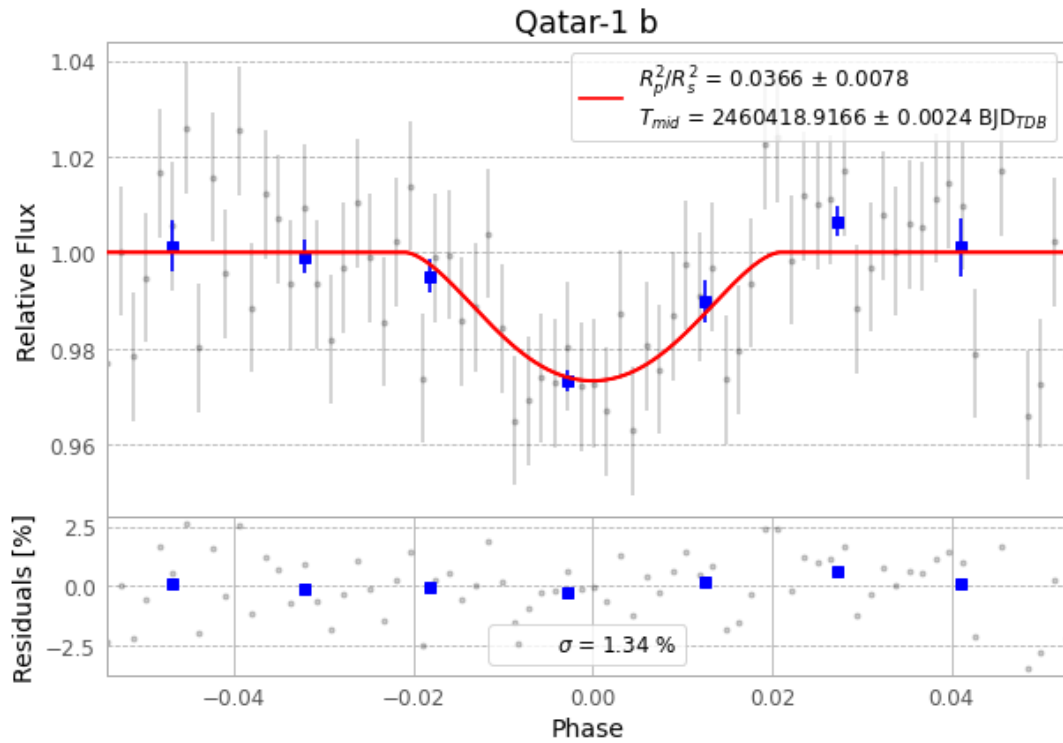


Figure 1 — FinalLightCurve_Qatar-1 b_2024-04-18.png (SHA-256 41da04b3113007e7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [320, 241], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 473acd7383fbc63b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

74 FITS frames (49 MB), Qatar-1240418080915.FITS → Qatar-1240418114815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-240418.log (ccf038fdedb3...), FinalLightCurve_Qatar-1 b_2024-04-18.png (41da04b31130...), FinalLightCurve_Qatar-1 b_2024-04-18.csv (4598e0f0749a...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 00:55Z.